

C-Zero Company Summary

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C-Zero is a small tech startup based in Santa Barbara, California, with the purpose of decarbonizing methane gas and turning it into hydrogen gas and solid, sequestered carbon. Started by Eric McFarland in 2018 with funding from Shell. They do this by taking a multiphase liquid thermocatalytic and running methane gas through it, allowing the multiphase liquid to separate the carbon from the methane through a process called catalyzed methane pyrolysis. As a convenient side effect, the sequestered carbon is immiscible in the liquid catalyst, and so falls out of solution, where it can be easily collected to be used or buried.

In the race to green energy, hydrogen fuel is a top contender, with various levels of “cleanness”, ranging from green to gold. C-Zero aims to gather “turquoise” hydrogen, which could be a precursor for the cleanest “gold” hydrogen. The company’s target is \$1.50 per kilogram of fuel. Currently, the price per kilogram of hydrogen is \$25, so the company has a long way to go, but also provides a very competitive price for the market.

Currently, the company is still a privately owned startup, meaning it has no stock price. Despite this, it has had several fundraising campaigns, the most recent of which was in June 16th, 2022, in which it raised 34 million dollars, bringing its total to just under 50 million in funding since it began fundraising back in 2020. It’s not a very large company, with its linkin page listing only 27 employees, and their website listing 24. In spite of their small numbers, many in the team are seasoned engineers and scientists, with years of experience, with a healthy group of recent graduates.

Their 2022 fundraising was for their primary expansion, the influx of capital allowing the company to build its first pilot plant, “which is expected to be online in Q1 2023” and will “be capable of producing up to 400 kg of hydrogen per day from natural gas with no CO₂

emissions.” (Czero.com) The primary financiers were led by SK Gas, a subsidiary of South Korea's second largest conglomerate, the SK Group. It is clear that the plant will be built in South Korea, a country that empties a tanker (which can hold up to 10 million cubic feet of liquid natural gas) every other day in the summer, and every 36 hours during the winter. It is clear that in the shift to clean energy, South Korea is searching for all alternatives to energy, and ramping up natural gas is only the start to the solution. C-zero clearly aims to be part of this expansion into cleaner energy.

There have been no recent expansions or large movements to speak of, though the company's news page was updated just two days ago as of this analysis. The team is active in their outreach efforts, going to conferences with Exxon and Monolith on Stanford's Natural Gas Initiative Methane Pyrolysis Panel, attending Expos around the country, and making their presence heard through a litany of news organizations. There has been little to no news on the plant, which presumably has started construction since its conception a year and a half ago.

In conclusion, C-Zero, a small startup based in California, has its sights set big, helping countries like South Korea take their increasing consumption of liquid natural gas and sequester the carbon before burning, allowing the country to continue down its path to a net zero future. Founded in 2018, it has already received funding and is in the process of building its pilot plant in South Korea. Led by a small but experienced and eager group of engineers and scientists, they have a lot of potential in the methane pyrolysis field, a field that is rapidly expanding and has large implications for energy security in the future of a world run by natural gas.

Sources

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